

HU Collection

TMUA Mock Set B Paper 1

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	E	C	B	C	A	C	C	C	A
Q	11	12	13	14	15	16	17	18	19	20
Ans	B	C	A	D	D	A	D	D	B	C

Each solution records the intended method and, where there is one, the trap the question was built around.

Question 1. Answer B.

Let $P = \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$ and multiply by $\sin \frac{\pi}{7}$. Folding three times with $2 \sin \theta \cos \theta = \sin 2\theta$,

$$\sin \frac{\pi}{7} P = \frac{1}{2} \sin \frac{2\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = \frac{1}{4} \sin \frac{4\pi}{7} \cos \frac{4\pi}{7} = \frac{1}{8} \sin \frac{8\pi}{7} = -\frac{1}{8} \sin \frac{\pi}{7},$$

so $P = -\frac{1}{8}$. Since $\cos \frac{3\pi}{7} = -\cos \frac{4\pi}{7}$, the required product is $\frac{1}{8}$. Option E is P itself, the answer to the question that was not asked; the sign flip on the last cosine is the whole point.

Question 2. Answer E.

The cubic is $(x-2)^3 - 1$, so put $u = x-2$; the limits become -3 and 3 and the integrand is $|u^3 - 1|$, which changes sign at $u = 1$. Then

$$\int_{-3}^1 (1 - u^3) du + \int_1^3 (u^3 - 1) du = 24 + 18 = 42.$$

Ignoring the modulus gives $\int_{-3}^3 (u^3 - 1) du = -6$, option A, which is negative and therefore impossible for the integral of a non-negative function. A negative answer to a modulus integral is always a signal to look again.

Question 3. Answer C.

The binomial coefficient $\binom{x}{j} = \frac{x(x-1)\cdots(x-j+1)}{j!}$ is a polynomial in x of degree j , and for a non-negative integer k we have $2^k = \sum_{j=0}^k \binom{k}{j}$, where the terms with $j > k$ vanish. So $p(x) = \sum_{j=0}^4 \binom{x}{j}$ is a quartic with $p(k) = 2^k$ for $k = 0, \dots, 4$, and it is the only one, two quartics agreeing at five points being equal. Then

$$p(5) = \sum_{j=0}^4 \binom{5}{j} = 2^5 - \binom{5}{5} = 31.$$

The same answer comes from a difference table: the fourth differences of $1, 2, 4, 8, 16$ are constant, and extending the table one step gives 31. The trap is 32, which assumes a polynomial can imitate 2^x one step further than its degree allows.

Question 4. Answer B.

Write $f(x) = x^3 - 3x + 1 = (x - a)(x - b)(x - c)$. Then $a^2 - 2 = (a - \sqrt{2})(a + \sqrt{2})$, and

$$\prod (a - \sqrt{2}) = -f(\sqrt{2}), \quad \prod (a + \sqrt{2}) = -f(-\sqrt{2}),$$

so the required product is $f(\sqrt{2})f(-\sqrt{2}) = (1 - \sqrt{2})(1 + \sqrt{2}) = -1$. No root is ever found.

The trigonometric route explains the answer. Putting $x = 2 \cos \theta$ turns $x^3 - 3x$ into $2 \cos 3\theta$, so the roots are $2 \cos \frac{2\pi}{9}$, $2 \cos \frac{4\pi}{9}$, $2 \cos \frac{8\pi}{9}$, and the map $x \mapsto x^2 - 2$ is $2 \cos \theta \mapsto 2 \cos 2\theta$, which permutes the three roots. The product is therefore $abc = -1$ by Vieta.

Question 5. Answer C.

Put $u = x + \frac{1}{x}$ and $v = y + \frac{1}{y}$. Then $u + v = 1 + \frac{1}{x} + \frac{1}{y} \geq 1 + \frac{4}{x+y} = 5$, the inequality being $\frac{1}{x} + \frac{1}{y} \geq \frac{4}{x+y}$ (AM–HM, or clear denominators to $(x - y)^2 \geq 0$). Then

$$u^2 + v^2 \geq \frac{(u + v)^2}{2} \geq \frac{25}{2},$$

with equality throughout at $x = y = \frac{1}{2}$, where each bracket is $\frac{5}{2}$. Substituting $y = 1 - x$ and differentiating is possible but ugly, and the two-step inequality is what the question is testing.

Question 6. Answer A.

Expanding,

$$(\sin x + 2 \cos x)(2 \sin x + \cos x) = 2 \sin^2 x + 5 \sin x \cos x + 2 \cos^2 x = 2 + 5 \sin x \cos x.$$

Now $\sin x \cos x$ ranges over $[-\frac{1}{2}, \frac{1}{2}]$ (it equals $\frac{1}{2} \sin 2x$; or note $(\sin x \pm \cos x)^2 \geq 0$ gives $|\sin x \cos x| \leq \frac{1}{2}$ with both ends attained). So $M = \frac{9}{2}$, $m = -\frac{1}{2}$ and $Mm = -\frac{9}{4}$.

Question 7. Answer C.

Partial fractions in n alone go nowhere, because of the 2^n . The trick is to split the numerator against the factor 2 rather than against n and $n + 1$: write $n + 2 = 2(n + 1) - n$, so

$$\frac{n + 2}{n(n + 1)2^n} = \frac{2}{n 2^n} - \frac{1}{(n + 1)2^n} = \frac{1}{n 2^{n-1}} - \frac{1}{(n + 1) 2^n}.$$

With $b_n = \frac{1}{n 2^{n-1}}$ this is $b_n - b_{n+1}$, so the partial sum is $b_1 - b_{N+1} = 1 - \frac{1}{(N + 1)2^N}$, and the infinite sum is 1. The first three partial sums are $\frac{3}{4}$, $\frac{11}{12}$, $\frac{31}{32}$, which is the pattern a candidate without the decomposition would have to guess from; option B is the first term on its own.

Question 8. Answer C.

Let $s = \sin x + \cos x$, so $\sin x \cos x = \frac{1}{2}(s^2 - 1)$. The equation becomes $s^2 - 1 = 2s - 2$, i.e. $(s - 1)^2 = 0$, so $\sin x + \cos x = 1$. Squaring, $\sin x \cos x = 0$, which gives the candidates $0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$; but squaring is not reversible, and checking in $\sin x + \cos x = 1$ kills 180° and 270° . The solutions are $0^\circ, 90^\circ, 360^\circ$ with sum 450° . Option F is the unchecked sum 900° ; option A forgets the endpoints.

Question 9. Answer C.

Label the heptagon $ABCDEFGH$, with side 1, short diagonal p (two steps apart, e.g. AC) and long diagonal q (three steps apart, e.g. AD). Every regular polygon is cyclic, so Ptolemy applies to any

four vertices. On the quadrilateral $ABCE$ the sides are $AB = BC = 1$, $CE = p$, $EA = q$ and the diagonals are $AC = p$, $BE = q$, so Ptolemy gives

$$AC \cdot BE = AB \cdot CE + BC \cdot EA, \quad pq = p + q,$$

so $\frac{1}{p} + \frac{1}{q} = 1$ on dividing by pq . The lengths themselves are $p = 2 \cos \frac{\pi}{7}$ and $q = 1 + p - \frac{1}{p}$, neither of which is needed. Ptolemy on $ABCD$ gives the companion identity $p^2 = 1 + q$.

Question 10. Answer A.

Since $\alpha\beta = -1$, we have $\frac{1}{\alpha} = -\beta$ and $\frac{1}{\beta} = -\alpha$, so the required value is $-(\alpha^5 + \beta^5)$. Writing $s_n = \alpha^n + \beta^n$ and using $s_{n+1} = 2s_n + s_{n-1}$ (from the equation), $s_1 = 2$, $s_2 = 6$, $s_3 = 14$, $s_4 = 34$, $s_5 = 82$. The answer is -82 . Option F is the sign slip.

Question 11. Answer B.

The tangent to $y = x^2$ at $x = a$ is $y = 2ax - a^2$; the tangent to the second curve at $x = b$ is $y = (6 - 2b)x + b^2 - 10$. Matching gradients and intercepts, $a = 3 - b$ and $-(3 - b)^2 = b^2 - 10$, which reduces to $2b^2 - 6b - 1 = 0$. The two common tangents correspond to the two roots b_1, b_2 , with gradients $6 - 2b_i$, so

$$(6 - 2b_1)(6 - 2b_2) = 36 - 12(b_1 + b_2) + 4b_1b_2 = 36 - 36 - 2 = -2$$

by Vieta, without solving for b at all. (The gradients are $3 \pm \sqrt{11}$.)

Question 12. Answer C.

The series is geometric with ratio $x - 1$; it converges exactly when $|x - 1| < 1$, i.e. $0 < x < 2$, and there its sum is $\frac{1}{1 - (x - 1)} = \frac{1}{2 - x}$. Outside that interval the formula $\frac{1}{2 - x}$ still makes sense as a number but the series diverges, which is the trap in option A. At $x = 0$ the terms are ± 1 and the series does not converge, so option D is also wrong.

Question 13. Answer A.

A chord of length 6 in a circle of radius 5 sits at distance $\sqrt{25 - 9} = 4$ from the centre. Any line through $(3, 1)$ has distance at most $|OP| = \sqrt{10} < 4$ from the origin, since the perpendicular distance to a line through P cannot exceed the distance to P itself. So no such chord exists. The instinct to answer 2 works for chords of length 8, which need distance $3 < \sqrt{10}$.

Question 14. Answer D.

Compute a few terms: $a_1 = 3$, $a_2 = \frac{1}{2}$, $a_3 = -\frac{1}{3}$, $a_4 = -2$, $a_5 = 3$. The sequence has period 4. (Structurally, if $g(x) = \frac{x-1}{x+1}$ then $g(g(x)) = -\frac{1}{x}$, so g has order 4.) The product over one period is $3 \cdot \frac{1}{2} \cdot (-\frac{1}{3}) \cdot (-2) = 1$. Since $2026 = 4 \cdot 506 + 2$, the product is $a_1 a_2 = \frac{3}{2}$.

Question 15. Answer D.

Let $t = 2^x + 2^{-x}$, so $t \geq 2$ by AM-GM, with $t = 2$ only at $x = 0$, and $4^x + 4^{-x} = t^2 - 2$. The equation becomes $t^2 - kt + 2 = 0$. Each root $t > 2$ gives two values of x (symmetric under $x \mapsto -x$), $t = 2$ gives one, and $t < 2$ gives none. The roots multiply to 2, so at most one of them can be ≥ 2 . Hence exactly two solutions in x needs exactly one root $t > 2$, i.e. the quadratic $h(t) = t^2 - kt + 2$

satisfies $h(2) < 0$, which is $6 - 2k < 0$, so $k > 3$. At $k = 3$ the roots are 2 and 1, giving only $x = 0$. Option B is the discriminant condition and ignores the constraint $t \geq 2$ entirely.

Question 16. Answer A.

$y' = 4x^3 - 12x^2 + 2ax = 2x(2x^2 - 6x + a)$. There is always the stationary point $x = 0$, so we need $2x^2 - 6x + a = 0$ to have no real roots: $36 - 8a < 0$, i.e. $a > \frac{9}{2}$. The boundary case $a = \frac{9}{2}$ gives the double root $x = \frac{3}{2}$, which is still a second stationary point (a stationary point of inflection), so option B is wrong. The factor $2x$ is the other trap: $x = 0$ is a stationary point for every a , so "no real roots" applies to the quadratic factor only.

Question 17. Answer D.

Split the integrand. The function $x\sqrt{4-x^2}$ is odd, so its integral over $[-2, 2]$ is 0. What remains is

$$2 \int_{-2}^2 \sqrt{4-x^2} dx,$$

and $y = \sqrt{4-x^2}$ is the upper half of the circle of radius 2, so the integral is the area of that semicircle, 2π . The answer is 4π . There is no need for an antiderivative of $\sqrt{4-x^2}$; recognising a definite integral as an area is the method. Option E takes the whole circle, option A forgets the even part.

Question 18. Answer D.

The angle at C is fixed, so C lies on an arc of a circle through A and B , and the sine rule in the circumcircle gives $2R = \frac{AB}{\sin 45^\circ} = \frac{4}{\sqrt{2}/2}$, so $R = 2\sqrt{2}$. The centre is at distance $\sqrt{R^2 - 2^2} = 2$ from AB . Since 45° is acute, C is on the major arc, on the far side of the centre from AB , and the largest height is $2 + 2\sqrt{2}$ at the midpoint of that arc. The maximum area is $\frac{1}{2} \cdot 4 \cdot (2 + 2\sqrt{2}) = 4 + 4\sqrt{2}$. Option A is the height, not the area.

Question 19. Answer B.

Substitute in each integral. In the first, $u = 2x$ gives $\frac{1}{2} \int_0^4 f = 3$, so $\int_0^4 f = 6$. In the second, $u = x + 1$ gives $\int_2^4 f = 5$. In the third, $u = 3 - 3x$ reverses the limits and contributes a factor $\frac{1}{3}$, giving $\frac{1}{3} \int_0^3 f = 2$, so $\int_0^3 f = 6$. Then

$$\int_3^4 f = \int_0^4 f - \int_0^3 f = 6 - 6 = 0.$$

The information is enough, so option G is wrong; the value 5 in option E is $\int_2^4 f$, read off carelessly.

Question 20. Answer C.

Do not try to invert f . Since f is increasing and $f(2) = 10$, the graph of f^{-1} on $[0, 10]$ is the reflection of the graph of f on $[0, 2]$ in the line $y = x$, and together the two regions fill the 2×10 rectangle:

$$\int_0^{10} f^{-1}(x) dx = 2 \cdot 10 - \int_0^2 (x^3 + x) dx = 20 - (4 + 2) = 14.$$

Option A is $\int_0^2 f$, the wrong piece of the rectangle.